10 . Smart Irrigation Powered by Artificial Intelligence (AI)

Harini and other .at.el 2024 published a paper in Smart irrigation systems revolutionizing farming. The system makes decisions using sensors that monitor humidity,temperature, and environment to gather data at the appropriate times. With the use of artificial intelligence algorithms like machine learning models .

The advantage of this approach is that it will support local knowledge combined with technology application to enhance agricultural production and environmental protection.

The project's hardware includes the Arduino Uno controller as its core, the DHT11 sensor which measures temperature and humidity, as well as the soil humidity sensor; two solenoid valves control the water flow.

The archive Survey of “Machine Learning countable Applications in Exactness Farming"examines the critical headways in advanced farming due to the integration of machine learning and fake insights. machine learning models in different perspectives of accuracy in agribusiness, including edit abdicate expectation, infection discovery, weed discovery, edit acknowledgment, trim quality evaluation, dribble water system, water quality administration, soil properties examination, climate determining, creature welfare, animal generation, collecting methods, and fertilizer suggestion.

Dataset It is a tree-structured classifier, where inside hubs speak to the highlights of a dataset, branches speak to the choice rules, and each leaf hub speaks to the outcome. In a choice tree, there are two hubs, which are the choice hub and leaf hub. The choices for the test are performed on the premise of the highlights of the given dataset. It is a graphical representation for getting all the conceivable arrangements for a problem or decision based on given conditions.

The results reveal a significant positive impact on water conservation and efficiency compared to conventional irrigation methods. Based on real-time data, leading to reduced water wastage and optimized resource utilization. Furthermore, the implementation of the smart irrigation system demonstrates tangible benefits in terms of crop yield and quality.

23. IoT with Evolutionary Algorithm Based Deep Learning for Smart IrrigationSystem

P. Suresh and others .at.el (2022) in Computers, Materials & Continua. The paper is called "IoT with Evolutionary Algorithm Based Deep Learning for Smart Irrigation System".

This smart irrigation system using ai checks the soil to see if it needs water. It uses sensors to check the soil moisture, soil temperature, air temperature and humidity. The system collects all this data to make decisions about the irrigation system using ai. The smart irrigation system using ai is really good at figuring out when the plants need water.

This means it uses water effectively and does not waste it. The smart irrigation system using ai does all this with help from people. People are involved in making sure the smart irrigation system using ai works properly. The DS18B20 sensor checks the soil temperature. The EC-1258 sensor checks the soil moisture. The system also uses a server that does three things. First it predicts what the environment will be like using deep support vector machine regression. Then it uses clustering to make sure the predictions are accurate. Finally it decides if the plants need water using an Artificial Immune Optimization Algorithm tuned deep belief network model for the smart irrigation system using ai.

The smart irrigation system using ai looks at the data from the sensors. The smart irrigation system using ai is very good at making these decisions.

The Autonomous smart irrigation system using ai is accurate. It helps to save water. The smart irrigation system using ai has an accuracy of 0.971, precision of 0.980, recall of 0.981 and F1-score of 0.980. This means the irrigation system using air reduces water wastage and uses resources in a way. The smart irrigation system using ai is really effective, in saving water and taking care of the plants.

11. Intelligent Irrigation System With Smart Farming Using ML and Artificial Intelligence Techniques

Manikandan P and others.at.el published a paper in Research Square in 2024. The paper is called "Intelligent Irrigation System With Smart Farming Using ML and Artificial Intelligence Techniques".

One good thing about the Intelligent Irrigation System is that it helps reduce the costs of farming. This is great for the economy because it helps the economy grow. It reduces the amount of work people have to do on the farm. This is very helpful for the farmers because they can focus on things.

In the paper the Intelligent Irrigation System checks how well it can monitor data from the internet of things using a Node-RED dashboard and Mosquitto broker. The Intelligent Irrigation System checks how well it can forecast crop yield using linear regression. This is important because it helps the Intelligent Irrigation System make decisions. The Intelligent Irrigation System can predict when it will rain and how water the crops will need.

The people behind the paper used dataset samples from the Tamil Nadu Rainfall Prediction dataset and observations from farm sensors. They did this to check the parameters and forecast how much water would be needed for the crops. he Intelligent Irrigation System can help farmers grow crops and make money.

The results of the Intelligent Irrigation System are really good. The Intelligent Irrigation System works better than machine learning models. The proposed ensemble learning algorithm, which is called SVM-KNN is really accurate. The Intelligent Irrigation System is correct 90 percent of the time. This is better than the SVM model, which's correct 72 percent of the time. The Intelligent Irrigation System is also better than the KNN model, which's correct 76 percent of the time.

7. Smart Irrigation System Using Artificial Intelligence and IOT

Chandra and others .at.al published a paper in 2025 about a Smart Irrigation System. This system uses Artificial Intelligence and Internet of Things technologies. The system also uses sensors to measure soil

moisture, temperature and humidity. It even uses cloud computing and special models like regression and neural networks.

The Smart Irrigation System works with data from sensors in the field and weather information from sources. This includes things like how likely it's to rain and what the temperature will be like. The system makes this data better by removing noise, stabilizing it and filling in missing parts. The results showed that the system is very good at predicting when the soil will lose moisture in weather and soil conditions. It can automatically turn valves on and off which saves a lot of water compared to doing it by hand.

However the Smart Irrigation System has some problems. It does not work well with crops that are growing at stages or with soil that has different profiles for multiple crops. The system also has trouble when the network is slow or not connected all the time because it relies much on the cloud. The system needs people to set thresholds by hand of being able to learn and adjust on its own. It also does not have ways to show what is happening in multiple fields like pictures or heatmaps.. It does not have ways for people to use voice commands or understand it in different languages, which makes it hard for farmers who are not good with technology to use the Smart Irrigation System.

12. Design and Implementation of an Integrated IoTand Artificial Intelligence System for Smart Irrigation Management

Prasad et al. (2025) about an irrigation system. This system uses intelligence and the internet of things to help date palm orchards in dry areas like Kebili, Tunisia.

A Southern Kebili experience refers to a typical Mediterranean climate, marked by hot, dry summers and mild, relatively wet winters .A yearly average temperature hovers around 20-25°C. A terrible summer's temperature peak exceeds 45°C. The topography of the area is relatively flat, with subtle . undulations. This even terrain impacts water distribution and drainage patterns, influencing the effectiveness of irrigation techniques. Infrastructure in the region primarily supports agricultural activities .

The researchers used a set of data that included real-time information about the environment and soil. They got this data from sensors placed near the roots and leaves of date palm trees in Tunisia.

When they looked at what was most important for irrigation they found that temperature and humidity were the factors. The smart irrigation system gave each date palm tree 150 liters of water per day which made the irrigation more efficient and used water overall. This saved between 30% and 60% water that would have been used. The date palm trees got the right amount of water.

This study only looked at date palm orchards in areas so the results might not work for other types of soil, climates or crops. Also the smart irrigation system needs to be connected to the internet. The system was designed with them in mind. The Prasad et al. Study shows that the smart irrigation system can be very useful for date palm orchards in regions, like Kebili, Tunisia and can help the date palm trees grow well.

01. Topic : Autonomous Crop Irrigation System using Artificial Intelligence

Savita Choudhary, Vipul Gaurav, Abhijeet Singh, and Susmit Agarwal (2019), in their paper titled "Autonomous Crop Irrigation System using Artificial Intelligence,
" published a paper in which they

combined multiple machine learning techniques for rainfall and water-requirement prediction — using Random Forest for rainfall classification/regression, ARIMA for rainfall time-series forecasting, and PLSR (Partial Least Squares Regression) for estimating crop water requirements, along with expert irrigation decisions.
On the hardware part, a Raspberry Pi combined with NodeMCU / IoT modules was used to control the irrigation system.
The dataset was sourced from Government of India rainfall records, with the initial deployment focused in the Punjab region, specifically wheat crops, along with soil and weather inputs.
For evaluation, the model used a 70/15/15 train/validation/test split, measuring both rainfall prediction accuracy and irrigation decision accuracy. In controlled-environment testing, the system achieved an overall accuracy of 80%, while for Punjab's annual rainfall prediction, the model predicted 545 mm against
an actual 549.5 mm, giving an accuracy of 83%. Based on these predictions along with real-time soil moisture data, the pump was automatically controlled without manual input.
However, the system's major limitation is its dependency on external weather-station data, and the authors suggest replacing this with on-premise sensors for better rural deployment, additionally, the
The model's generalization to other crops and regions requires further adaptation.
The key strength of this work lies in its strong autonomy , the system independently predicts rainfall and water needs and decides pump ON/OFF status without requiring human intervention.

04. AI-Driven Smart Irrigation System Using Edge-Based Embedded Controllers

Manthila.at.el published a paper about "AI-Driven Smart Irrigation System Using Edge-Based Embedded

Controllers" on year 2026, in which they used edge-based TinyML techniques, running on a Raspberry Pi 4 for on-device AI inference and an STM32 microcontroller for real-time actuation, allowing fully local decision-making without cloud dependence. The system uses environmental and soil-moisture data - temperature, humidity, and rainfall along with crop-specific thresholds to decide irrigation timing.
The dataset was collected from semi-arid agricultural field-test conditions, based on real local climatic and soil-moisture measurements rather than a public dataset. The results showed strong performance--decision accuracy over 94%, inference time of 34 ms, up to 38% water-use reduction, 20% increase in crop yield, and power consumption under 2.5W - making it efficient for real-world, resource-limited deployment. The main limitation is that the model needs retraining for different crops or climates, as it was trained specifically on semi-arid conditions. Future work proposed includes reinforcement learning, federated learning, and solar power integration. The key strength is its strong fit for autonomous irrigation — combining low-latency, low-power, on-device AI with direct actuation, without needing internet connectivity.

16. Topic : Implementation of Smart Irrigation using IoT and Artificial Intelligence

Tace Y.; Elfilali S.; Tabaa M.; Leghris C.at.el,"Implementation of Smart Irrigation using IoT and Artificial Intelligence" 2023, published a paper in which they used supervised machine learning models — KNN, SVM,and Naive Bayes for irrigation pump decisions, with Arduino handling sensor acquisition, Node-RED for monitoring and data flow, and MongoDB Cloud for storage.
The models were trained using pump-decision labels to automate irrigation control.
The dataset consisted of soil humidity, air humidity, temperature, and rain sensor readings, collected from a controlled-environment plant setup rather than an external public dataset.
The results showed that KNN performed the best among the compared models, with an accuracy of 98.4%with a very low RMSE of 0.016, outperforming both SVM and Naive Bayes in sensor readings for irrigation decisions. As for limitations, the authors note the need for larger and more diverse databases, with testing additional algorithms to strengthen the model's reliability. They also propose semi-supervised learning as a promising direction for future work, because manually labeled data can be limited in real deployments. The key strength lies in its simple, direct workflow , mapping raw sensor values straight to pump ON/OFF decisions through a lightweight ML pipeline, while comparing multiple models (KNN, SVM, Naive Bayes) to identify the best-performing one. This makes the system easy to replicate and interpret, especially for smaller-scale or prototype irrigation setups.

27. Topic : Smart Irrigation System Based on IoT and Machine Learning

Some researchers, Y. Tace, M. Tabaa, S. Elfilali, C. Leghris, H. Bensag and E. Renault .at.al about Smart Irrigation System Based on IoT and Machine Learning. In this paper they used machine learning models

like KNN, Neural Network, Naïve Bayes, SVM and Logistic Regression to decide when to irrigate. They even built a web application to monitor and visualize the Smart Irrigation System Based on IoT and Machine Learning.
The data for the Smart Irrigation System Based on IoT and Machine Learning was collected over months from a special place that was made to help plants grow. This place had sensors to measure the moisture in the soil, the temperature, the humidity and the rain. The researchers did not use data from people; they collected their own data.
The results of the Smart Irrigation System Based on IoT and Machine Learning showed that KNN was the model. It was very accurate with an accuracy of 98.3%. The other models, like Neural Network, Naïve Bayes, SVM and Logistic Regression were not as good as KNN.

The researchers said that they need data and more algorithms to make the Smart Irrigation System Based on IoT and Machine Learning better. They also said that the data they collected is not good enough for other researchers to use.

The good thing about this paper is that it is simple and easy to understand. The researchers used the data from the sensors to decide when to turn the pump on or off. They also made a web application to monitor the Smart Irrigation System Based on IoT and Machine Learning which makes it easy to use in the world.

29. Topic : A Smart Irrigation Control System That Leverages Machine Learning

M. Karthikaa; J. Steffi et al. published a research paper about "A Smart Irrigation Control System That Leverages Machine Learning" on 2025, in which they used a Random Forest classifier for predicting /dry soil status, combined with IoT temperature and soil-moisture sensors connected via ESP32.
To handle class imbalance in the data (dry readings than wet) they applied SMOTE, ADASYN and Mix-up techniques alongside cross-validation. A FastAPI backend was used for prediction and automatic pump ON/OFF control.
The dataset was a real-world soil dataset containing temperature (in °C) moisture (in %) and soil status (Wet=1/Dry=0) readings, which naturally had dry samples rather than wet ones.
The results showed improvement through imbalance-handling techniques: the baseline Random Forest achieved 84% accuracy, which improved with SMOTE (89%) ADASYN (91%) and finally Mix-up gave the best result at 92% accuracy 91% precision, 92% recall and 91% F1-score.
As a limitation the paper does not report size or broader multi-field validation meaning the results are based on a fairly limited testing scope. The authors propose improvements including weather forecasting, LSTM/GRU time-series prediction, mobile-app integration and solar-powered operation.
The key strength of this paper is its handling of class imbalance using multiple techniques (SMOTE, ADASYN, Mix-up), with Mix-up giving the strongest results. Directly connected to automatic pump control making it both accurate and practically deployable